

PBL PROJECT PROPOSAL REPORT

Smart Electricity Management System
Using Data Structures and Algorithms

Graphic Era Deemed to be University

B.Tech CSE (AI & ML)

Semester III

DSA with C/C++

1. PROJECT AND TEAM INFORMATION

Project Title:

Smart Electricity Management System Using Data Structures and Algorithms

Team Name:

Tech Titans

Team ID:

DSCPP-III-2026-T127

Mentor:

Ms. Shrishti Chamoli

Team Members

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2. PROPOSAL DESCRIPTION

Motivation

Electricity is used every day by appliances like air conditioners, fans, washing machines, air heaters, and computers, but people don't realize how much power they are using until they get their electricity bill. Users often don't know which appliance uses the most energy or when electricity is more expensive.

The goal of this project is to help users manage their electricity usage better. They only need to provide the name of the appliance, when they started using it, and when they stopped. Other details like the appliance's power rating, category, priority, and usage schedule can be pulled from a database.

The system can also show how much energy was used, estimate the cost, and tell when the appliance was used during high or low cost periods. It can send alerts if an appliance is used at an inconvenient time and suggest better times to use it. This will show users how much they could save.

This project demonstrates how C++ and Data Structures and Algorithms can be applied to solve a real-world problem in a structured and useful way.

Current Solutions

Electricity usage is tracked using traditional meters, smart meters, utility apps, and home monitoring systems. These tools can show total usage and billing details, but they do not always explain how to make changes.

A user might know their bill is high, but not which appliance is causing it or if moving the appliance's usage time could save money. Our system goes beyond just showing data. It combines detailed energy calculations, cost estimates, peak and off-peak analysis, alerts, and smart scheduling into one application.

All the technical details about appliances are stored in a database, so the user does not have to enter them manually.

Project Goals and Milestones

The main aim is to create a C++ based Smart Electricity Management System that helps users better understand and control their appliance usage.

The main goals are:

- Maintain a database of common appliances.
- Allow users to input only the appliance name, start time, and end time.
- Calculate energy use and cost.
- Identify peak and off-peak times for each appliance.
- Send alerts for high-cost usage.
- Suggest better times to use appliances.
- Provide energy-saving ideas and what-if savings.
- Generate energy reports and budget alerts.

The project will include several milestones: requirement analysis, database design, C++ class development, implementation of Data Structures and Algorithms, energy and cost calculations, smart scheduling, recommendation logic, testing, and final system integration.

Project Approach

The system will be developed using C++ with Object-Oriented Programming and Data Structures and Algorithms.

A predefined database will store information like appliance ID, name, category, power, usage, priority, scheduling, and standby power. Users will choose an appliance and input its start and end times. The system will retrieve the appliance's profile and calculate the duration, energy use, and cost.

C++ classes and objects will be used to represent appliances. Encapsulation will keep the data organized, while inheritance, polymorphism, and abstraction can be used for different types of appliances.

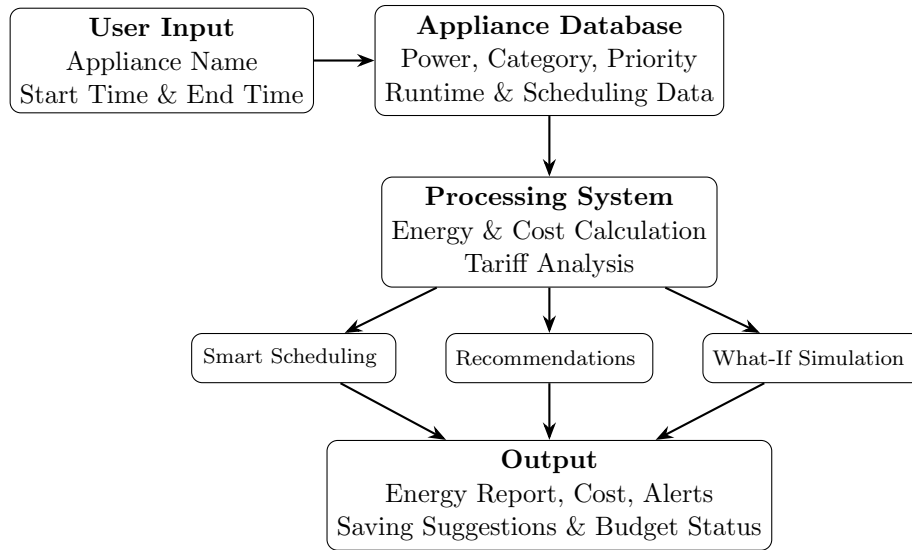
STL containers like vectors, maps, queues, and priority queues will be used based on their purposes. Searching and sorting algorithms will help analyze usage records.

The system will compare the entered time with predefined tariff periods. If the appliance is used during a high-cost period, an alert and an alternative time will be suggested. File handling will be used to store usage records and generate reports.

3. SYSTEM ARCHITECTURE AND OUTCOME

System Architecture (High-Level Diagram)

The system follows a simple flow from user input to data processing and then to useful results.



System Flow:

Appliance Name, Start Time, End Time → Smart Scheduling → Appliance Database → Power, Category, Priority, Runtime and Scheduling Data → Processing System → Energy & Cost Calculation → Tariff Analysis → Recommendations → What-If Simulation → Energy Report, Cost, Alerts, Saving Suggestions and Budget Status

Project Outcome / Deliverables

The final deliverable will be a working C++ based Smart Electricity Management System.

The system will include a set of pre-set data about typical home appliances, along with details like how much power they use, what category they fall into, their priority level, and when they should be scheduled.

The user will provide the name of the appliance, when they want it to start and when they want it to stop. The system will then figure out how much energy the appliance uses and how much it might cost. It will also check if the usage is during a busy or less busy time and give warnings when needed.

The system will offer smart scheduling ideas, tips to save energy, pretend scenarios to see what might happen, reports for each appliance, and alerts when the budget is getting close. To handle all this data smoothly, the system will use data structures like vectors, maps, queues, and priority queues.

The final results will include a working C++ program, the source code, test cases, examples of outputs, a description of the system's design, and all the project details.

Assumptions

The system assumes that there are already set profiles for appliances and information about how much energy is charged at different times.

Instead of using real-time data from a smart meter or sensors, the system will use the times the user provides to show how much energy is being used. Because of this, the cost estimates will be based on the stored appliance info and the selected pricing plan.

References

1. Uttarakhand Electricity Regulatory Commission (UERC), *Tariff Order of UPCL for FY 2026–27*.
2. Uttarakhand Electricity Regulatory Commission (UERC), *Tariff Orders and Regulatory Documents*.
3. C++ Standard Library documentation for STL containers, algorithms, file handling and exception handling.
4. Standard reference material on Data Structures and Algorithms using C++.